4 **(a)** gradient = E

y-intercept = − (R_F + r)

(b) $$R = \frac{\rho l}{A} = \frac{1.7 \times 10^{-8} \times 5.0}{\pi\left(\frac{0.26}{2} \times 10^{-3}\right)^2} = 1.6\ \Omega$$

The cable is made of 24 wires in parallel.

Resistance of cable = $\left(\frac{1}{1.6} \times 24\right)^{-1} = 0.067\ \Omega$

(c) As temperature increases, amplitude of vibration of lattice ions / metal ions (do not accept: metal atoms / molecules / particles) increase. Drift velocity of the free electrons decreases. No or insignificant increase in the number of charge carriers. Hence, overall effect resistance increases.

5 **(a)** Binding energy per nucleon is a maximum at around A = 56. Products splitting a Fe-56